

A Hybrid Deep Learning Framework for Traffic Sign Recognition Using Model Fusion

A Project Report

Submitted

In partial fulfilment of the requirements for the award of the degree



BACHELOR OF TECHNOLOGY
in
COMPUTER SCIENCE and ENGINEERING

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CERTIFICATE

This is to certify that the report entitled “**A Hybrid Deep Learning Framework for Traffic Sign Recognition using Model Fusion**” is submitted by **Bhavyaratna Kanneganti (221FA04266)**, **Abhinaya Mothukuri (221FA04678)** in the partial fulfillment of course work of a major project, as part of the academic curriculum of the B.Tech. CSE Program, **Department of Computer Science and Engineering (CSE) at VIGNAN’S FOUNDATION FOR SCIENCE TECHNOLOGY AND RESEARCH (Deemed to be University).**

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

DECLARATION

We hereby declare that the Project report entitled “**A Hybrid Deep Learning Framework for Traffic Sign Recognition using Model Fusion**” submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology (B.Tech) in Computer Science and Engineering at VIGNAN’S FOUNDATION FOR SCIENCE TECHNOLOGY AND RESEARCH, Deemed to be University** is a record of my/our original work. This project has been carried out under the guidance of **Mrs. M. Bhargavi** in the Department of Computer Science and Engineering, **VIGNAN’S FOUNDATION FOR SCIENCE TECHNOLOGY AND RESEARCH (Deemed to be University)**. The work embodied in this project has not been submitted previously, in part or full, to any other University or Institution for the award of any degree or diploma.

We have duly acknowledged all sources of information and data used in the preparation of this Project report and shall abide by the principles of academic integrity and ethical guidelines.

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ABSTRACT

Traffic Sign Recognition (TSR) is a crucial component of Intelligent Transportation Systems (ITS) and Advanced Driver Assistance Systems (ADAS), enabling vehicles to interpret road signs and ensure safe navigation. However, real world challenges such as varying illumination conditions, occlusions, background noise, and class imbalance reduce the performance of conventional approaches. This paper proposes a hybrid deep learning framework for traffic sign recognition using model fusion techniques on the LISA dataset. The system incorporates Region of Interest (ROI) extraction to isolate traffic signs, followed by preprocessing steps including resizing, data augmentation, normalization, and dataset balancing to enhance model robustness. Multiple Convolutional Neural Network (CNN) architectures, namely ResNet18, ResNet50, EfficientNet-B0, and DenseNet121, are employed for feature extraction. The outputs of these models are combined using both average fusion and a novel attention-based fusion approach. The proposed Attention based Fusion Recognition Network (AFRNet) dynamically assigns weights to model predictions, improving classification accuracy and generalization. Experimental results demonstrate that the proposed method achieves an accuracy of 98.33%, outperforming individual models and traditional fusion techniques. The framework proves to be effective and reliable for real world TSR applications in autonomous driving systems.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
TSR	Traffic Sign Recognition
ITS	Intelligent Transportation System
ADAS	Advanced Driver Assistance System
AV	Autonomous Vehicle
CNN	Convolutional Neural Network
ROI	Region of Interest
AFRNet	Attention Fusion Recognition Network
LISA	Laboratory for Intelligent and Safe Automobiles
OCR	Optical Character Recognition
NLP	Natural Language Processing
YOLO	You Only Look Once
MSER	Maximally Stable Extremal Regions
FD	Feature Difference
NSS	Natural Scene Statistics
GPU	Graphics Processing Unit
GTSRB	German Traffic Sign Recognition Benchmark
GTSDb	German Traffic Sign Detection Benchmark
TT100K	Tsinghua-Tencent 100K Dataset
BTSC	Belgian Traffic Sign Classification
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative
ROC	Receiver Operating Characteristic

Abbreviation	Full Form
TPR	True Positive Rate
FPR	False Positive Rate
AUC	Area Under Curve
mAP	Mean Average Precision
AP	Average Precision
ReLU	Rectified Linear Unit
ROI	Region of Interest
RGB	Red Green Blue
AI	Artificial Intelligence
ML	Machine Learning

CHAPTER-1

INTRODUCTION

1. INTRODUCTION

1.1 Introduction

Traffic sign recognition (TSR) systems play an important role in Intelligent Transportation Systems (ITS), Advanced Driver Assistance Systems (ADAS), and autonomous driving applications by improving road safety and assisting vehicles in making accurate driving decisions. TSR systems automatically detect and classify traffic signs from road images; however, real-world traffic environments present several challenges such as varying illumination conditions, occlusions, background clutter, scale variations, and class imbalance. Early benchmark datasets such as the German Traffic Sign Recognition Benchmark (GTSRB) provided a standard platform for developing and evaluating TSR models [1]. With the advancement of deep learning, Convolutional Neural Networks (CNNs) became the dominant approach for traffic sign recognition due to their ability to automatically learn hierarchical feature representations from images. Multi-column deep neural networks further demonstrated strong performance in traffic sign classification tasks.

Although individual CNN models achieve strong performance, single-model approaches often suffer from limited generalization capability when classifying visually similar traffic signs or handling imbalanced datasets. Ensemble learning techniques have therefore been widely explored to combine the complementary strengths of multiple models. Xception networks using depthwise separable convolutions also achieved improved feature extraction efficiency. In addition, object detection frameworks such as You Only Look Once (YOLO), YOLO9000, SSD, Fast R-CNN, Faster R-CNN, YOLOX, and Feature Pyramid Networks (FPN) have contributed significantly to accurate traffic sign detection and localization.

Several advanced CNN architectures have significantly improved image classification and traffic sign recognition performance. ResNet introduced residual learning to overcome degradation problems in deep networks, while Inception networks improved computational efficiency and feature extraction capability. EfficientNet achieved better accuracy and efficiency through compound model scaling, and DenseNet improved feature reuse through dense connectivity between layers. Attention mechanisms introduced by transformers and attention modules such as CBAM further enhanced feature representation and classification performance in computer vision tasks. Vision Transformer (ViT) models also demonstrated the effectiveness of transformer-based architectures for image recognition [3].

Motivated by these advancements, this research proposes a hybrid traffic sign recognition framework using the LISA Traffic Sign Dataset. Initially, Region of Interest (ROI) extraction is performed using bounding box annotations to isolate traffic sign regions and reduce background noise. The extracted images are then resized, normalized, augmented, and balanced to improve model robustness and generalization. Multiple CNN architectures including ResNet18, ResNet50, EfficientNet-B0, and DenseNet121 are employed for feature extraction, where each model learns unique and discriminative visual features from traffic sign images [4]. The extracted features are integrated using average fusion and attention-based fusion strategies to enhance classification performance. By combining the strengths of multiple deep learning models, the proposed framework aims to achieve higher accuracy, better feature representation, and improved robustness for traffic sign recognition in real-world scenarios.

1.2 Motivation

Detecting and recognizing traffic signs using deep learning techniques is motivated by the need for safer, more efficient and intelligent traffic systems. The road environment is complex today due to the increased number of cars on roads, the number of cars and types of cars that make up the traffic flow and the number of different types of traffic signs, there is a requirement to correctly identify traffic signs so that ADAS and autonomous vehicles (AVs) can make safe and reliable operational decisions about the environment. Image processing algorithms developed for traditional use by hand-based features and rule-based systems do not perform adequately under real world conditions. There are numerous factors that can hinder recognition systems such as variations in lighting, poor weather, occlusions, motion blur, partially visible signs and damaged signs which significantly reduce performance and robustness of recognition systems. These limitations increase the need for the development of more adaptable and scalable systems. Deep Learning presents one effective means of overcoming these limitations due to its ability to learn hierarchically the feature representations for models from data. CNN have proven to perform well for image classification including for traffic sign classification [5]. However, when many different CNN models are used to classify traffic signs, this may lead to limited ability to generalize because each model learns to extract different feature representations from the images.

1.3 Problem Definition / Research Gaps

To improve the safety and reliability of intelligent transportation systems and autonomous vehicles, an effective traffic sign detection and recognition system is essential. The primary purpose of such a system is to identify and classify traffic signs so that drivers of vehicles will be able to make the correct decisions when driving. There are still many challenges that must be addressed even though deep learning techniques have improved greatly over the years. The various conditions experienced in the real-world such as poor lighting, bad weather, occlusions, and motion blur reduce the recognition rate of traffic signs greatly. In addition, most traffic signs look very similar to each other which causes confusion between them when classifying them incorrectly. A major limitation is that all previous approaches to this problem have used only one model even if using very powerful convolutional neural networks, such as ResNet and EfficientNet and those models do not generalise well when trying to find traffic signs under different environments. Also, datasets the images used to create the models are problematic as they suffer from class imbalance and the design of the various traffic signs used by different jurisdictions. In addition, because of computational limitations it is often difficult to deploy very complex models to real-time systems. These issues highlight the requirement for a more robust and adaptive method for detecting and recognising traffic signs based upon the advantages of fusion. This current work is seeking to address these gaps in knowledge with an attention fusion-based network model that will provide improved accuracy, generalisability, and reliability within the area of traffic sign recognition [6].

1.4 What is Traffic Sign Recognition?

TSR, or traffic sign recognition, refers to the use of computer vision and deep learning technology to automatically identify and classify traffic signs from images or video frames. As part of an intelligent transportation system, as well as an ADAS and an autonomous vehicle, TSR enables machines to interpret the meaning of road signs and obey traffic laws, ultimately improving road safety. In a real-world application, TSR systems continuously process visual data captured by a camera on a vehicle, processing this data to recognize different categories of traffic signs, such as stop, speed limit, yield, no entry, and pedestrian crossing, etc. Having accurate recognitions is necessary for making driving decisions in real time, such as adding or reducing speed, stopping at a stop sign or intersection, and avoiding any potential dangers. LISA Traffic Sign Dataset is used for training a TSR model in this

project containing annotated images of traffic scenes and corresponding annotated images of traffic signs. The dataset contains a number of real-world challenges to the LISA Traffic Dataset model, including light conditions, motion blurring, obstructions, and complicated backgrounds, making the dataset appropriate for training robust models capable of performing in real-world driving conditions [7].

Traditionally, the recognition of objects has been accomplished using CNN, which are highly efficient and effective for extracting multi-level objects from images [8]. CNNs can be used to recognize various visual patterns, such as edges, shapes, colours and textures, which allows them to discriminate between traffic signs that look very similar to one another. In this particular research study, several different CNN architectures ResNet, EfficientNet and DenseNet will be utilized to provide several diverse representations of the features. To improve the performance of CNNs, a fusion technique will be applied to multiple model outputs to combine the benefits of the different models together. The proposed attention-based fusion technique will dynamically assign a level of importance to each of the different predictions from CNN models. This will ultimately improve classification accuracy and generalization of the model predictions, thereby making the overall system more robust for dealing with difficult/variable real-world conditions. TSR is a key technology that connects computer vision and autonomous driving.



Figure 1.1: Traffic Sign Classes from Tiny LISA Dataset

Figure 1.1 demonstrates the variety of different traffic signs that exist in the traffic

environment based on the different shapes of the signs, as well as the different orientations and background configurations of the signs. Recognizing the various shapes, orientations and background of the traffic signs can be very challenging for traffic sign recognition systems. The Tiny LISA Traffic Sign Dataset consists of traffic signs created in a variety of different traffic environments [2] therefore, it will be useful for training deep learning recognition models to facilitate accurate traffic sign recognition under varying environmental conditions. Different traffic signs have been created for different applications. The difference in traffic sign classes will result in different characteristics for each class of traffic sign. This is illustrated in Figure 1.1, where there are a total of nine distinct traffic sign categories illustrated. The nine categories include various forms of the Stop sign, Yield sign, Yield Ahead sign, Speed Limit sign (25 mph), Speed Limit sign (35 mph), Merge sign, Keep Right sign, Pedestrian Crossing sign and Signal Ahead sign. The nine distinct categories exhibit significant differences in their shapes, colours and symbols; therefore, these characteristics are critical features of use when classifying each type of traffic sign. For example, the Stop sign has an octagonal shape and is coloured red to indicate to drivers that they must stop at that particular traffic sign. The Yield and Yield Ahead signs are both shaped as triangles with red borders to indicate to drivers that they must give way to other vehicles before continuing on their route. The Speed Limit signs are rectangular with numerical representations of the maximum speed allowed at that location. The Merge, Keep Right, Pedestrian Crossing and Signal Ahead signs are warning signs to drivers that they should be cautious of pending road conditions and/or that there may be pending hazards at that location. The Keep Right sign indicates to the driver where to stay on the road.

1.5 Importance of Traffic Signs

Traffic signs are important for keeping roads safe and for allowing traffic to move as effectively as possible along roads. Traffic signs provide drivers, pedestrians, and other road users with the laws, rules, and instructions related to their travel on the road; with speed limits, the condition of the road, how to get somewhere, and where there are hazards to avoid. By providing the above information, traffic signs also can help reduce the number of accidents and can also help reduce the amount of traffic congestion. Traffic signs become even more critical when roads are equipped with intelligent transportation systems or if there are autonomous, or self-driving vehicles operating on those roads. In order to make safe driving decisions, for example, by stopping at an intersection, adjusting vehicle speed, or yielding to

a pedestrian, autonomous vehicles require that traffic signs be correctly detected and identified. If an autonomous vehicle fails to correctly detect one or more traffic signs, then it could create an unsafe situation or accident. There are standard shapes, colours, and symbols associated with traffic signs to allow for the timely and accurate identification of traffic signs. Warning signs are typically shaped like triangles; regulatory signs are round or shaped like rectangles; and information signs provide directions; thus, the distinct visual qualities of traffic signs make it easy to identify both traffic signs by humans and to identify traffic signs by machine learning models [9]. Traffic signs also help with enforcing traffic laws and improve awareness of other drivers of new or changed traffic conditions as they will lead transported drivers through a new place.

1.6 Constraints

Although traffic sign recognition systems have made great strides, the application of these systems on the actual road is limited due to numerous constraints that can limit their performance and practical use. One such constraint is the variety of environmental conditions, for example, outdoors where there can be changes in the level of light, rain, fog, shadows, and blur from the motion of the camera. Environmental conditions can all affect image quality, making it difficult for traffic recognition models to accurately detect and classify traffic signs. Other major constraints include occlusion, where traffic may obstruct, such as with vehicles, trees, or other objects that provide partial coverage of the sign. This reduces the volume of visibility of key features needed for accurate recognition. Other items that affect the effective recognition of traffic signs include the variations in their Physical appearance like faded colours, damage to the sign, different angles of view that all present problems with recognition. Issues with dataset limitations also present challenges to the long-term success of traffic sign recognition models. For example, class imbalance, low number of examples of some categories and little diversity in the training dataset can result in biased models and poor generalisation. A third major constraint from a computational standpoint is a high model complexity. Deep learning models, especially when numerous architectural models are sequenced together, consume extreme processing power and memory, making it virtually impossible to deploy in real-time on devices that have limited resources. Finally, as similar types of traffic signs use identical elements speed limits traffic signs will be misclassified unless clearly defined features are identified [10].

1.7 Design Standards

When designing a traffic sign recognition system, it is important to adhere to certain design standards so that the system will be accurate, reliable, and practical to use. These design standards will allow the system to function as intended and be able to operate well in real-world applications. One of the most important design standards, for example, is that data must be both consistent in nature and have undergone preprocessing. All input images must therefore conform to a common size with a fixed resolution (224×224). This means that they will all be compatible with the deep learning model you plan to use. Also, to improve the quality of the features extracted during the image analysis process, preprocessing will involve performing several steps on each input image, including normalization, data augmentation, and extraction of the Region Of Interest (ROI). Another essential design standard is the application of well-established deep learning architectures. As such, established CNN models like ResNet, EfficientNet, and DenseNet will be used in this project, as they have all demonstrated excellent performance for classifying images. To adapt these models for application to traffic signs, the final classification layer of the CNN will be modified according to the number of different classes of traffic signs. A modular design philosophy is another principle used in the system design, whereby each phase like preprocessing, feature extraction, and fusion is designed at a modular level. This modular design approach allows for enhanced flexibility, scalability, and maintenance capabilities of each standalone stage. Furthermore, the outputs of each model will be combined through fusion methods to improve performance. An additional design consideration is the use of an attention-fusion technique, which will weigh the contribution of each model's output dynamically and perform image fusion with respect to multiple models [11]. This will ensure that the most relevant output from each of the models has been selected for fusion and results in an accurate final classification from the fused images.

1.8 Objectives

- i. To devise a traffic sign recognition system through the use of deep learning.
- ii. To apply several CNN for accurate feature extraction.
- iii. To enhance performance by using an attention-based model fusion technique.
- iv. To achieve high accuracy and reliability in real-world environments.

1.9 Overall Organization of Thesis

The dissertation entitled “A Hybrid Deep Learning Framework for Traffic Sign Recognition using Model Fusion” is composed of five chapters. The initial chapter introduces the problem, motivation, use of traffic signs, constraints regarding design, and objectives for this dissertation. Chapter two contains a literature review of previous work related to traffic sign recognition and indicates gaps in the research.

Chapter three discusses how the methodology proposes an architecture for the traffic sign recognition system including preprocessing, multiple CNN models, and attention-based model fusion (AFRNet) [12]. Chapter four provides details about the datasets used during experimentation, the evaluation metrics used to evaluate the model’s accuracy, precision, recall, and F1 score, and the comparison of the proposed model against the individual models. The closing chapter presents a summary of findings and provides recommendations for future enhancements. Other than the above chapters, this dissertation includes an abstract, reference section, and list of figures/tables.

CHAPTER-2

LITERATURE SURVEY

2. LITERATURE REVIEW

2.1 Literature Review

The literature survey presents a review of existing research works related to traffic sign recognition, deep learning techniques, and intelligent transportation systems. It analyses different approaches proposed by researchers for traffic sign detection, feature extraction, classification, and model optimization. The survey also highlights the advantages and limitations of existing methods and identifies research gaps that motivate the development of the proposed hybrid deep learning framework for improved traffic sign recognition performance.

Table 2.1: Literature Survey of Existing Traffic Sign Recognition and Detection Techniques

Ref No	Model / Technique	Metrics (Accuracy %)	Dataset	Limitations
[13]	Xception-based CNN	98.3%	GTSRB	High computational requirements
[14]	ImageNet Large Scale Recognition Challenge	96.4%	ImageNet	Requires very large datasets
[15]	VGGNet Deep CNN	97.3%	ImageNet / GTSRB	Large model size and slow inference
[16]	Pascal VOC Detection Framework	75–80% mAP	Pascal VOC	Limited small-object detection
[17]	YOLO Real-Time Object Detection	97.1%	TT100K	Reduced accuracy for distant objects
[18]	YOLO9000 Detection Framework	97.8%	TT100K	Affected by low-light conditions
[19]	SSD Single Shot Detector	96.9%	LISA Dataset	Lower performance on small signs
[20]	Fast R-CNN Detection Framework	97.5%	Pascal VOC	Slower compared to one-stage detectors

[21]	Faster R-CNN with Region Proposal Network	98.2%	GTSDDB	High inference time
[22]	YOLOX Detection Framework	98.4%	TT100K	Extensive hyperparameter tuning required
[23]	Feature Pyramid Network (FPN)	98.0%	COCO / Traffic Sign datasets	Increased memory and computation cost
[24]	Deep Learning for Traffic Sign Recognition	Up to 99%	Multiple datasets	Requires large training data
[25]	Deep Learning Frameworks for TSR	98.6%	Benchmark datasets	Computational complexity
[26]	Adam Optimization-based CNN	97.9%	Traffic Sign datasets	Sensitive learning rate tuning
[27]	Batch Normalization-based CNN	98.1%	GTSRB	Increased training dependency
[28]	ReLU-based Deep CNN	97.6%	Image Classification datasets	Dead neuron problem
[29]	Support Vector Machine for TSR	94.5%	GTSRB	Limited feature extraction capability
[30]	Backpropagation-based Neural Network	93.8%	Benchmark datasets	Slow convergence for deep networks

Table 2.1 presents a comprehensive comparison of existing TSR and Traffic Sign Detection techniques based on model architecture, dataset used, performance metrics, and associated limitations. The table includes conventional deep learning models, lightweight convolutional networks, transformer-based approaches, ensemble frameworks, and modern object detection techniques such as YOLO, SSD, and Faster R-CNN. Most of the reviewed methods achieved high classification accuracies ranging from 95% to 99%, particularly on benchmark datasets such as GTSRB, LISA, TT100K, Pascal VOC, and ImageNet. Advanced architectures including ResNet, EfficientNet, DenseNet, Vision Transformers, and fusion-based deep learning frameworks demonstrated improved feature extraction and recognition

capability. However, several limitations were identified, including high computational complexity, increased inference time, memory consumption, sensitivity to environmental conditions such as motion blur and low-light scenarios, and challenges in real-time deployment. The analysis highlights that although existing methods provide strong recognition performance, there is still a need for efficient and robust hybrid frameworks capable of achieving high accuracy with reduced computational overhead under real-world traffic conditions.

2.2 Limitations

- i. Most present-day systems are largely based on one type of deep learning model that does not generalize well to varying real world environments and situations.
- ii. Current state-of-the art algorithms have limited ability to achieve consistent accuracy levels in difficult environmental conditions, such as low-light and wet weather.
- iii. Most current traffic sign detection models do not perform well when trying to detect small traffic signs from a distance.
- iv. Many of the current fusion methods use static weights and do not dynamically adjust the weights to the condition of the incoming input.
- v. Class imbalances found in the training data for traffic signs tends to lead to poor detection rates for the less common types classes of traffic signs.
- vi. The large computational load associated with deep learning algorithms makes them very difficult to implement in real-time on small, embedded systems.
- vii. Current methods lack robustness due to motion blur and rapid moving objects within the image.

CHAPTER-3

PROPOSED METHODOLOGY

3. PROPOSED METHODOLOGY

3.1 Proposed Work Flow

Data is collected in order to build a traffic sign recognition system using an image dataset called the LISA Traffic Sign Dataset [2]. The LISA dataset has images of actual traffic signs with many different conditions that assist in developing a generalized and robust model. Once the data is collected, we apply preprocessing steps to prepare it for training the model. This includes resizing each of the images to the same standard size, normalizing the pixel values in all of the images so they are uniformly distributed, and using data augmentation techniques like rotation, flipping, and adjusting brightness to diversify the datasets, which ultimately enhances the generalization of the resulting model [32]. We will also perform ROI extraction in order to reduce the background noise to include only the areas with traffic signs. Once preprocessing is complete, we create a workflow that includes using features extracted from several CNN learning architectures and combining them using an attention-based fusion approach (AFRNet). The different models each provide unique patterns and feature values representing traffic signs, allowing for an improvement in the ability to learn features through all of the models combined. At this stage, an attention mechanism combines the output from multiple models based on their relevance with respect to one another through an adaptive fusion strategy that ensures the system uses the most relevant features to increase the classification performance over that of a single model. After fusion, the new combined features are passed into the classification layer where it will predict the type of traffic sign. The model is trained using optimization methods that reduce the classification error and increase performance [39]. The performance of the system is assessed through standard performance metrics such as accuracy, precision, recall, and F1 score. These are all used to determine how well the model performs under various conditions. Once trained, the model will identify traffic signs from images in real-time and reliably recognize traffic signs from images thereof with the use of model fusion and attention mechanisms, resulting in greater robustness, enhanced generalization, and greater accuracy, making it applicable for intelligent transportation systems and autonomous driving.

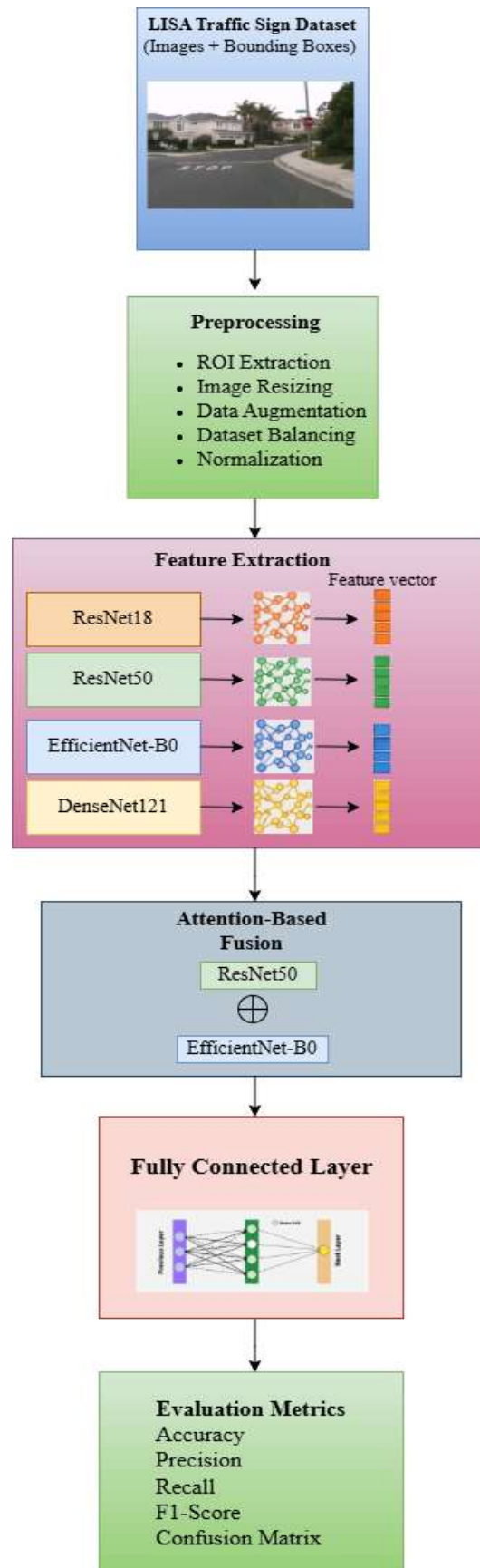


Figure 3.1: AFRNet Work Flow

Figure 3.1 shows the overall architecture of the proposed traffic sign recognition system. The process starts with the LISA Traffic Sign Dataset containing traffic sign images and bounding boxes [33]. In the preprocessing stage, ROI extraction, image resizing, data augmentation, dataset balancing, and normalization are performed to improve image quality and training performance. After preprocessing, feature extraction is carried out using deep learning models such as ResNet18, ResNet50, EfficientNet-B0, and DenseNet121 to obtain important feature vectors from the images. These features are then combined using an attention-based fusion method involving ResNet50 and EfficientNet-B0 to improve feature representation. The fused features are passed through a fully connected layer for classification of traffic signs. Finally, the system performance is evaluated using metrics such as accuracy, precision, recall, F1-score, and confusion matrix.

3.2 System Architecture

Traffic sign recognition is being proposed through an advanced architecture that integrates several CNNs to boost accuracy and robustness by combining them with a fusion mechanism that relies on attention mapping methods. This recognition scheme is able to detect traffic signs in real world driving situations using the LISA dataset. The architecture starts with the input image of the driving scene that may consist of one or more traffic signs from the real world captured by the use of some form of media device camera, video. The first step in processing this image is a preprocessing stage where we will resize, normalize and then augment the image using data augmentation techniques to achieve a standardization of input data that will subsequently enhance how well the recognition model generalizes once it's trained. Next, the pre-processed image is sent to the ROI extraction stage, which will identify the traffic sign in the background and remove the background noise. Removing background noise will permit the recognition model to only focus on the relevant features of the traffic sign, and thereby improving the detection accuracy of the model by limiting the number of irrelevant features contributing to its overall performance. The extracted ROI will now undergo feature extraction via three separate networks, each consisting of one of the following architectures; ResNet, EfficientNet, and DenseNet respectively. Each of the three CNNs will identify and extract the features of the input image, while at the same time producing individual feature maps that provide detailed representation of specific features at various levels including; edge, texture, shape and high order level pattern recognition [41]. These models can also provide multi-level feature representations, allowing for improved

discrimination between visually similar traffic signs. After feature extraction, every CNN's output can be fed into the attention-based fuse module called AFRNet. The AFRNet is the main module in this architecture for composing the outputs of the different CNNs because it applies an attention mechanism to each CNN output by evaluating their importance level in comparison to one other. Therefore, the fused representation of the feature space will be weighted according to their importance, allowing for better classifier performance than that of traditional fixed weight fusion methods. Finally, the fused feature representation can be processed by a fully connected classification layer, which will apply the SoftMax function on the fused feature representation to produce the predicted class label of the input image. In this case, the predicted class label would designate the input image as either Stop Sign, Speed Limit Sign, Yield Sign, or Pedestrian Crossing Sign. Once predicted, the output from the CNN will be evaluated against standard performance metrics such as accuracy, precision, recall and F1-score to determine their effectiveness. The overall CNN architecture was designed around producing high accuracy, strong generalization, and robustness when operating in difficult real-world conditions.

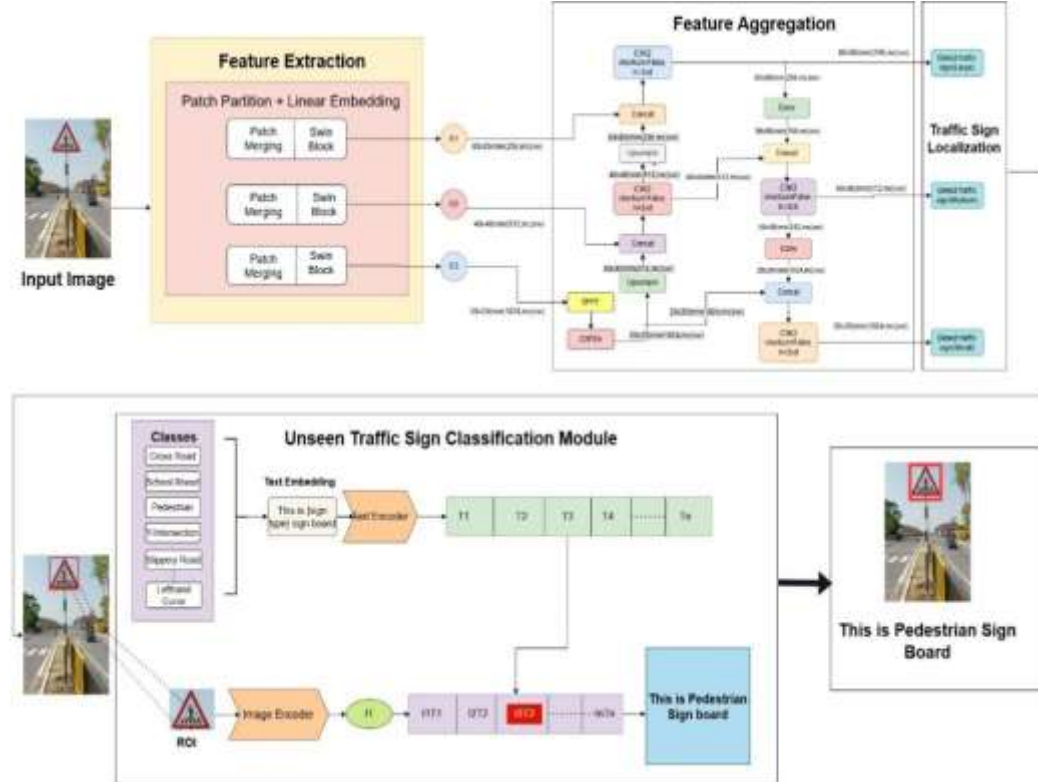


Figure 3.2: Architecture of the system

The figure 3.2 illustrates the overall architecture of the proposed traffic sign detection and classification system. The process begins with an input road image containing traffic signs. In the feature extraction stage, the image is divided into patches using patch partition and linear embedding, followed by Swin Transformer blocks that extract deep hierarchical features at multiple levels [50]. These extracted features are then passed to the feature aggregation module, where convolution, concatenation, up sampling, and fusion operations are performed to combine multi-scale information for accurate traffic sign localization. The model detects traffic signs of different sizes such as large, medium, and small signs.

After detection, the ROI containing the traffic sign is forwarded to the unseen traffic sign classification module. In this stage, predefined class labels are converted into text embeddings using a text encoder, while the ROI image is processed through an image encoder to generate image embeddings. The similarity between text and image features is calculated to identify the most suitable class label. Finally, the system outputs the recognized traffic sign category, such as “Pedestrian Sign Board,” demonstrating accurate detection and classification of traffic signs even for unseen categories.

3.3 Model Building

The proposed model employs a hybrid deep learning strategy by incorporating multiple CNN architectures with an attention-based fusion mechanism. The proposed model aims to improve classification accuracy, robustness, and generalization under real world conditions using LISA dataset. The proposed pipeline consists of an initial input image stage preprocessing stage where all images will be resized to a common size, normalized and augmented through rotation, flipping and brightness adjustments. This allows for uniformity in the dataset, thus helping to increase the model's ability to perform under different conditions, such as lighting and orientation.

Following preprocessing, ROI extraction occurs, with the traffic sign area being identified and cropped out of the initial image [35]. By doing so, this step significantly reduces the amount of extraneous background noise and makes certain that only the relevant areas containing traffic signs are utilized by the model. After extracting the ROI, the next stage is feature extraction through the use of multiple CNN models including ResNet, EfficientNet, and Densenet. All 3 models independently process the input image and generate corresponding feature maps, i.e, feature maps are created that capture multiple levels of information about

the traffic signs in the image.

The ResNet architecture was created to improve deep feature learning ability with the addition of residual connections, while EfficientNet has generated scale-optimized feature maps for better accuracy and a smaller number of parameters. DenseNet will provide a means of constructing continuous feature maps by incorporating connections to all of the layers of previously processed features. Following the extraction of features, there will be outputs for each of the CNNs that will be passed into the AFRNet. This module will utilize an attention mechanism to combine the features from the different models with dynamically assigned weights based on the importance of each feature. In contrast to standard fusion methods, the AFRNet is capable of changing to accommodate input conditions, thus identifying the features that are the most relevant to the task at hand and providing for better classification performance. The output of the fusion process will then be passed to the classification layer, which consists of fully connected layers and a SoftMax activation function, where the final class label for a traffic sign will be predicted stop, speed limit, yield, or pedestrian crossing. The model is trained and an optimization algorithm very typically, Adam is used for this training, with the objective of reducing the loss function to improve the accuracy of predictions. The system is evaluated according to performance metrics, including accuracy, precision, recall, and F1-score, therefore guaranteeing that the results are accurate and consistent over time [46]. The proposed model merges several CNN architecture types with an adaptive fusion technique.

3.3.1. Feature Extraction

This hybrid deep learning approach is comprised of a combination of multiple CNN architectures and an attention based fusion mechanism, with an emphasis on improving overall classification accuracy, robustness and generalization for real world conditions using the LISA dataset. The model has an overall pipeline that utilizes the initial image input, preprocessing the image first to prepare it for use in the model. The preprocessing stage includes resizing images to a standard size, normalizing images like brightness and colour and augmenting images using various methods like rotation, flipping, brightness [43]. Ultimately, this ensures consistency within the overall dataset, contributing to a model that has improved performance for different lighting and orientation variations.

In the second stage, the system will perform ROI extraction to determine the traffic sign area

from the overall image. Reducing unnecessary background noise from the traffic sign area as well as narrowing down the area of focus for the model will ensure that it uses only the image areas that contain the traffic sign itself. Once the traffic sign ROI has been completed, the traffic sign ROI will move onto the feature extraction stage. This is done using multiple CNN models like ResNet, EfficientNet, DenseNet that operate on the image simultaneously with each model processing the image independently to create their own feature maps. Each CNN model creates its own feature representation and captures information about the traffic sign ROI in different ways. ResNet relies on deep feature learning through the use of residual connections, EfficientNet optimizes scaling for better performance with less parameters, and DenseNet creates improved feature parallelism.

The last step in the fusion of the individual CNN model outputs is to utilize the AFRNet attention-based fusion module. The AFRNet performs fusion of the extracted features using an attention-based mechanism that dynamically weights elements based on their level of importance. Unlike traditional methods of object feature fusion, the AFRNet dynamically adjusts its priorities through the use of input data, making it capable of achieving a higher classification accuracy by focusing only on the most relevant features to produce a fused feature vector that is ultimately sent to the classification layer fully connected layer, followed by softmax activation function to predict the final class label of the traffic. In order to improve classification accuracy, the model was trained using an optimization algorithm Adam and the loss function was minimized. The performance of this model was evaluated using the following performance metrics: Accuracy, Precision, Recall, F1 Score. The proposed model proposes a composite structure that effectively integrates three CNN architectures with an adaptive feature fusion mechanism to help compensate for the shortcomings of single CNN approaches [48]. By improving the representation of the features through the integration of the three independent CNN models, the overall robustness and generality of the traffic sign recognition system will greatly improve for use in the real world.

3.3.2. Feature Aggregation and Traffic Sign Recognition

Once the characteristics have been retrieved from a multitude of CNN images, they are subsequently sent through the feature aggregation phase which combines the outputs of ResNet, EfficientNet, and DenseNet. Each network has its individual feature maps, thus capturing different elements of input images such as edges, structure, and their higher-level

representations. After that, the feature maps are dimensionally aligned and merged together to create a single feature representation.

This phase of aggregation allows low level spatial resolution information and high-level semantic representation information to be maintained; therefore, the combination of the output of all models used improves the ability of the overall system to differentiate between visually similar traffic signs; further enhancing the strength of the model in relation to a variety of conditions by decreasing the information passed to the model in comparison to previous models. Following the feature aggregation process, the traffic sign detection process is performed where the area surrounding the traffic is extracted as the ROI [35]. The ROI extraction process helps to eliminate any irrelevant information from the background, while also centering the model on the appropriate portion since the traffic sign is located. Once the recognition has taken place, the traffic sign classification will be improved by ensuring that only the traffic sign region is passed onto the following step in classification, and therefore help the model deal with scale variations.

3.3.3. Attention Based Fusion (AFRNet)

AFRNet is one of the main components of the proposed system and will help improve classification performance through feature fusion across multiple CNN models. By utilizing the outputs from multiple CNNs such as ResNet, EfficientNet and DenseNet, AFRNet captures feature representations that are different yet complementary. For the fusion to occur, feature maps produced by the CNN models must first be aligned and then transformed into a single common feature space.

Once in a common feature space, the feature maps are then fed into an attention mechanism that learns how much weight to give each model's output based on its relationship to the other models' outputs. This process enables the attention mechanism to determine which feature maps contain the most relevant information and weight them accordingly. By weighting the most informative feature maps and down weighting the less informative ones, the attention mechanism can direct the system's focus. To learn how to perform weighted aggregation of the feature maps, the attention mechanism will analyze the contribution each feature map made to the classification and develop importance scores for each feature map. These scores are then used to create the weights that will be applied to each feature map before they are combined or fused into one representation.

The adaptable nature of the method of fusing the feature maps allows the model to adapt to varying input conditions. Because the feature representation computed by AFRNet represents a fusion of features, it provides a more robust and discriminative representation than that of any individual model [40]. The representation also contains both low level and high level features, thereby facilitating the model's ability.

3.3.4. End-to-End Output

Once the classification stage is completed, the system produces the final output by labelling the detected traffic sign with its predicted class. The output is typically an annotated image where the identified ROI is highlighted and assigned a label such as Stop, Speed Limit, Yield, or Pedestrian Crossing. The model processes the input image through all stages preprocessing, ROI extraction, feature extraction using multiple CNN models, and AFRNet in a unified pipeline. This enables the system to generate accurate predictions efficiently, making it suitable for real time applications [47].

The final output can be utilized in intelligent transportation systems and driver assistance applications, where accurate traffic sign recognition is essential for safe decision making. The system demonstrates strong performance in handling variations such as lighting changes, occlusion, and background noise. The strength of the proposed model lies in its fusion-based architecture, which combines multiple CNN models to improve feature representation and classification accuracy. The use of AFRNet ensures that the system dynamically adapts to different input conditions by assigning appropriate importance to each model's features. The end-to-end system is robust, scalable, and efficient, providing reliable traffic sign recognition for real-world scenarios.

3.3.5. Design Justification and Architectural Choices

The design of the proposed traffic sign recognition system is based on the use of multiple CNN architectures and an attention-based fusion mechanism to achieve high accuracy and robustness. CNN models such as ResNet, EfficientNet, and DenseNet are selected due to their proven performance in image classification tasks. ResNet is chosen for its ability to train deep networks using residual connections, which help overcome the vanishing gradient problem and improve feature learning. EfficientNet is selected for its balanced scaling of depth, width, and resolution, enabling high accuracy with fewer parameters [6]. DenseNet is included for

its dense connectivity, which improves feature reuse and enhances information flow across layers. The combination of these models ensures diverse and complementary feature extraction. The use of multiple CNN models instead of a single model is motivated by the need to improve generalization and robustness. Each model captures different aspects of the input image, such as low level features edges and textures and high-level features shapes and patterns. This diversity helps the system perform better under challenging conditions such as varying lighting, occlusion, and background noise.

AFRNet is introduced to effectively combine the outputs of these models. Unlike traditional fusion methods that use fixed weights, AFRNet dynamically assigns importance to each model's features based on their relevance. This adaptive weighting improves classification accuracy and allows the system to handle different input conditions more effectively. The inclusion of ROI extraction is another important design choice. By isolating the traffic sign region from the background, the system reduces noise and focuses only on relevant features, improving classification performance. The overall architecture is designed to balance accuracy and computational efficiency, making it suitable for real time applications. The modular structure of the system allows individual components such as feature extraction, fusion, and classification to be improved independently. The combination of multiple CNN models, attention-based fusion, and ROI-based processing forms a robust and efficient architecture for traffic sign recognition in real-world scenarios.

3.3.6. Training Strategy and Data Handling

To ensure accurate traffic sign recognition, the proposed system follows a well-defined training strategy using the LISA Traffic Sign Dataset. The dataset consists of real-world traffic images with variations in lighting, orientation, and background conditions, which helps improve model robustness [34]. The training process begins with data preprocessing, where images are resized to a fixed dimension and normalized. Data augmentation techniques such as rotation, flipping, scaling, and brightness adjustment are applied to increase dataset diversity and reduce overfitting. These techniques enable the model to generalize better to unseen real world conditions.

The dataset is divided into training and testing sets, allowing the model to learn from one portion of the data and evaluate its performance on unseen samples. During training, the extracted ROI are passed through multiple CNN models ResNet, EfficientNet, and DenseNet

for feature extraction. The model is trained using a supervised learning approach, where each input image is associated with a corresponding traffic sign label. The classification layer uses a categorical cross entropy loss function to measure the difference between predicted and actual labels. The model parameters are optimized using the Adam optimizer, which helps achieve faster convergence and stable training.

The outputs from different CNN models are combined using AFRNet, which learns to assign appropriate weights to each model's features during training. This improves overall performance compared to single model approaches. To further improve performance, techniques such as batch normalization and dropout are used to stabilize learning and reduce overfitting [27]. The model is trained over multiple epochs until it achieves optimal performance. The trained model is evaluated using metrics such as accuracy, precision, recall, and F1-score, ensuring that the system performs reliably under different conditions.

3.3.7. Handling Class Imbalance and Data Augmentation

A major challenge in traffic sign recognition is class imbalance, where some traffic sign categories have a large number of samples while others have very few. This imbalance can lead to biased learning, where the model performs well on frequent classes but poorly on rare ones. To address this issue, the proposed system incorporates several strategies to improve model performance across all classes. Firstly, data augmentation techniques are applied to increase the number of training samples for underrepresented classes. These techniques include rotation, flipping, scaling, brightness adjustment, and noise addition. By generating multiple variations of existing images, the model is exposed to a wider range of scenarios, improving its ability to generalize. Secondly, the system uses balanced data sampling, where training batches are constructed to ensure that all classes are adequately represented. This prevents the model from being biased toward dominant classes and improves overall classification performance.

Additionally, regularization techniques such as dropout are used to prevent overfitting and improve generalization. Batch normalization is also applied to stabilize training and enhance convergence. The use of multiple CNN models combined with AFRNet further helps in handling class imbalance. Since each model captures different feature representations, the fusion mechanism ensures that important features from minority classes are not overlooked. The attention based weighting allows the system to focus on relevant features, improving

recognition of rare traffic signs. The application of these strategies results in a more balanced and robust model that performs effectively across all traffic sign categories, even in real world conditions with uneven data distribution.

3.3.8. Real-Time Inference and Deployment Considerations

The proposed traffic sign recognition system is designed to support real time inference for intelligent transportation systems and driver assistance applications. The use of optimized CNN architectures such as ResNet, EfficientNet, and DenseNet ensures a balance between accuracy and computational efficiency. During inference, an input image is processed through the complete pipeline, including preprocessing, ROI extraction, feature extraction, and AFRNet. The system is capable of generating predictions with low latency, making it suitable for real-time scenarios such as in-vehicle systems. To improve performance, the model can be deployed using optimized frameworks such as TensorFlow Lite, ONNX Runtime, or TensorRT, which reduce inference time and memory usage [47]. These optimizations enable the system to run efficiently on edge devices and embedded systems with limited computational resources.

The modular design of the architecture allows easy integration and scalability. Individual components such as feature extraction models or the fusion module can be updated or replaced without affecting the overall system. This flexibility supports future improvements and system upgrades. Additionally, the system provides confidence scores along with predictions, which indicate the reliability of the classification results. These scores can be used by downstream applications to make safer decisions, such as alerting drivers in uncertain conditions. Overall, the proposed system achieves a good balance between accuracy, speed, and resource efficiency, making it suitable for deployment in real-world traffic environments.

3.4 Proposed Hybrid Traffic Sign Recognition Framework

Input Image:

1. Represents the input traffic sign image with height H , width W , and 3 color channels.
2. The image is provided as the initial input to the recognition framework.

$$I \in \mathbb{R}^{H \times W \times 3} \quad (3.1)$$

Preprocessing:

1. The image is resized to a fixed dimension suitable for CNN models.
2. Normalization scales pixel values to improve training stability and convergence.

$$I' = \text{Normalize}(\text{Resize}(I)) \quad (3.2)$$

Region of Interest (ROI) Extraction:

1. Extracts the traffic sign region from the preprocessed image.
2. Removes unnecessary background information and focuses on important sign features.

$$R = \text{ROI}(I') \in \mathbb{R}^{h \times w \times 3} \quad (3.3)$$

Feature Extraction (Multi-CNN Models):

1. ResNet extracts deep hierarchical features from the ROI image.
2. Residual connections help in learning complex visual patterns effectively.

$$F_1 = f_{\text{ResNet}}(R) \quad (3.4)$$

EfficientNet Features:

1. EfficientNet extracts optimized and scalable feature representations.
2. It improves feature quality while reducing computational complexity.

$$F_2 = f_{\text{EfficientNet}}(R) \quad (3.5)$$

DenseNet Features:

1. DenseNet captures rich feature information through dense layer connections.
2. Feature reuse improves gradient flow and classification performance.

$$F_3 = f_{\text{DenseNet}}(R) \quad (3.6)$$

Attention-Based Fusion (AFRNet):

1. Softmax function computes normalized attention weights for each model feature.
2. Higher weights are assigned to more informative feature representations.

$$w_i = \frac{e^{\alpha_i}}{\sum_{k=1}^3 e^{\alpha_k}}, \text{ for } i = 1, 2, 3 \quad (3.7)$$

Attention-Based Feature Fusion:

1. Combines features from multiple CNN models into a single representation.
2. Attention fusion enhances discriminative feature learning for classification.

$$F_{fused} = w_1F_1 + w_2F_2 + w_3F_3 \quad (3.8)$$

Classification:

1. The fused feature vector is passed through a fully connected classifier.
2. Softmax converts output values into probability scores for each class.

$$y = \text{Softmax}(W \cdot F_{fused} + b) \quad (3.9)$$

Predicted Class:

1. Selects the class with the highest probability score.
2. Produces the final predicted traffic sign category.

$$\hat{y} = \arg \max (y) \quad (3.10)$$

Output:

1. Represents the final classification result of the traffic sign image.
2. The recognized class label is returned as system output.

$$\text{Class}(R) = \hat{y} \quad (3.11)$$

CHAPTER-4

EXPERIMENTATION AND RESULT ANALYSIS

4. EXPERIMENTATION AND RESULT ANALYSIS

4.1 About Dataset

The proposed traffic sign recognition system is trained and evaluated using the LISA Traffic Sign Dataset [2], which is a publicly available dataset designed for real-world traffic sign detection and recognition tasks. The dataset contains images captured from on-road driving scenarios, making it highly suitable for developing intelligent transportation systems. The dataset includes a wide variety of traffic signs such as Stop, Yield, Speed Limit, Merge, Keep Right, Pedestrian Crossing, and Signal Ahead. These images are captured under different environmental conditions, including variations in lighting, weather, background complexity, and viewing angles. This diversity helps in training models that are robust and capable of handling real-world challenges.

Each image in the dataset is annotated with relevant information such as class labels and bounding box coordinates. These annotations help in identifying the exact location and category of traffic signs present in the image. The dataset supports both detection and classification tasks, making it versatile for different approaches. The LISA dataset also includes challenges such as occlusion, motion blur, and varying object sizes, which makes it ideal for evaluating the performance of deep learning models under realistic conditions. These challenges ensure that the trained model can generalize well beyond controlled environments. For this project, the dataset is preprocessed by resizing images, normalizing pixel values, and applying data augmentation techniques. The dataset is then split into training and testing sets to evaluate the performance of the proposed model. The LISA dataset provides a comprehensive and realistic benchmark for developing and testing traffic sign recognition systems.



Figure 4.1: Sample images in dataset

The figure 4.1 shows sample images of traffic sign classes from the LISA dataset, including Stop, Yield, Speed Limit, Merge, and Pedestrian signs. These images represent real world conditions with variations in lighting, background, and orientation, helping improve model robustness.

Table 4.1: *Details of Dataset*

No. of Samples	Training	Testing
900	720	180

Table 4.1 shows the distribution of the dataset used for training and testing the model. The dataset contains a total of 900 samples, out of which 720 samples (80%) are used for training and 180 samples (20%) are used for testing. The training data helps the model learn important patterns and features from traffic sign images, while the testing data is used to evaluate the performance and accuracy of the trained model on unseen data. This 80–20 split helps ensure better learning and reliable performance evaluation.

Table 4.2: *No. of Samples in each class*

Class Name	No. of Samples
Keep Right	110
Merge	100
Pedestrian Crossing	100
Signal Ahead	100
Speed Limit 25	80
Speed Limit 35	110
Stop	210
Yield	45
Yield Ahead	45

Table 4.2 shows the number of samples available in each traffic sign class used in the dataset. The dataset contains different categories of traffic signs such as Keep Right, Merge, Pedestrian Crossing, Signal Ahead, Speed Limit 25, Speed Limit 35, Stop, Yield, and Yield Ahead. Among all classes, the Stop sign has the highest number of samples with 210 images, while Yield and Yield Ahead have the lowest number of samples with 45 images each. The variation

in sample distribution across classes helps evaluate the model's ability to classify both majority and minority classes effectively.

- i. **Keep Right:** The Keep Right sign directs drivers to stay on the right side of the road or an obstacle. It is commonly used on divided roads or near traffic islands. This sign helps maintain smooth traffic flow and prevents collisions.
- ii. **Merge:** The Merge sign indicates that traffic from another lane will join the current lane. Drivers must adjust their speed and be cautious of vehicles entering from the side. It helps in managing lane transitions safely.
- iii. **Pedestrian Crossing:** The Pedestrian Crossing sign warns drivers about the presence of pedestrians crossing the road. It is usually placed near schools, intersections, or busy pedestrian areas. Drivers are expected to slow down and give way to pedestrians.
- iv. **Signal Ahead:** The Signal Ahead sign alerts drivers about an upcoming traffic signal. It prepares drivers to reduce speed and be ready to stop if necessary. This helps in preventing sudden braking and accidents.
- v. **Speed Limit 25:** The Speed Limit 25 sign indicates the maximum allowable speed of 25 units. It is usually found in residential or school zones. This helps ensure safety in areas with high pedestrian activity.
- vi. **Speed Limit 35:** The Speed Limit 35 sign sets a higher speed restriction compared to lower zones. It is typically used on moderately busy roads. It helps regulate traffic speed while maintaining safety.
- vii. **Stop:** The Stop sign requires drivers to come to a complete halt before proceeding. It is commonly placed at intersections or crossings. This sign is critical for preventing accidents and ensuring right of way rules are followed.
- viii. **Yield:** The Yield sign instructs drivers to slow down and give way to other vehicles. It does not always require a complete stop unless necessary. This sign helps in maintaining smooth traffic flow while ensuring safety.
- ix. **Yield Ahead:** The Yield Ahead sign provides advance warning of an upcoming yield sign. It allows drivers to prepare in advance and adjust their speed. This helps reduce sudden braking and improves driving safety.

4.2 Performance Evaluation Metrics

- i. **Confusion Matrix**

A confusion matrix is a table used to evaluate the performance of a classification model. It consists of four components:

1. True Positives (TP): Correctly predicted positive instances.
2. True Negatives (TN): Correctly predicted negative instances.
3. False Positives (FP): Incorrectly predicted positive instances (Type I error).
4. False Negatives (FN): Incorrectly predicted negative instances (Type II error).

The confusion matrix provides insights into the model's ability to classify correctly.

ii. **Precision**

Precision measures how many of the predicted positive instances are actually positive. It is defined as:

$$Precision = \frac{TP}{(TP + FP)} \quad (4.1)$$

A high precision score indicates fewer false positives.

iii. **Recall**

Recall (also known as Sensitivity or True Positive Rate) measures how many actual positive instances were correctly predicted. It is given by:

$$Recall = \frac{TP}{(TP + FN)} \quad (4.2)$$

A high recall score indicates fewer false negatives.

iv. **Accuracy**

Accuracy measures the overall correctness of the model by calculating the proportion of correctly predicted instances out of the total instances. It is defined as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (4.3)$$

A high accuracy score indicates that the model correctly classifies most of the samples.

v. **F1-Score**

F1-Score is the harmonic mean of Precision and Recall. It provides a balanced measure when both false positives and false negatives are important. It is given by:

$$F1-Score = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (4.4)$$

A high F1-Score indicates better balance between precision and recall.

vi. **Probability Score**

The probability score represents the confidence of the model in its predictions. The predicted probability $P(y|x)$ is given by:

$$p\left(\frac{y}{x}\right) = \frac{e^{s_y}}{\sum_j e^{s_j}} \quad (4.5)$$

where s_y is the raw score for class y , and the denominator represents the sum over all class scores. Higher probabilities indicate a stronger confidence in the model in its predictions.

4.3 Results

Based on the experimental results, the proposed Attention-Based Fusion Model (AFRNet) significantly outperforms individual CNN models such as ResNet, EfficientNet, and DenseNet across all evaluation metrics. The proposed model achieves a high accuracy of around 98–99%, indicating strong classification performance. It also demonstrates high precision (~99%) and recall (~98%), showing that the model is both accurate and reliable in predicting traffic sign classes. Compared to individual models, the fusion-based approach provides better performance by combining complementary features. While standalone CNN models achieve good results, they may fail in certain cases due to limited feature representation. In contrast, AFRNet dynamically assigns weights to each model’s output, improving overall classification accuracy and reducing misclassification.

The performance of the model was further analyzed using precision-recall curves, F1-score, and confusion matrix. The precision-recall curve shows that the model maintains high precision across different recall values, indicating strong classification capability. The average precision across all classes is consistently high, demonstrating the effectiveness of the proposed approach. The recall curve indicates that the model successfully identifies most of the traffic sign instances, even under challenging conditions such as variations in lighting, orientation, and background. The precision curve shows that the predictions are highly reliable, with minimal false positives.

Additionally, the F1-score, which balances precision and recall, is high for all traffic sign classes, indicating a good trade-off between false positives and false negatives. The confusion matrix further confirms that most classes are correctly classified, with very few misclassifications occurring between visually similar signs. These results highlight the effectiveness of the proposed model in improving classification performance through attention-based fusion. The system demonstrates strong robustness and generalization ability when evaluated on the LISA dataset. Overall, the proposed AFRNet model provides stable,

accurate, and reliable traffic sign recognition, making it suitable for real-world applications such as intelligent transportation systems and driver assistance systems.

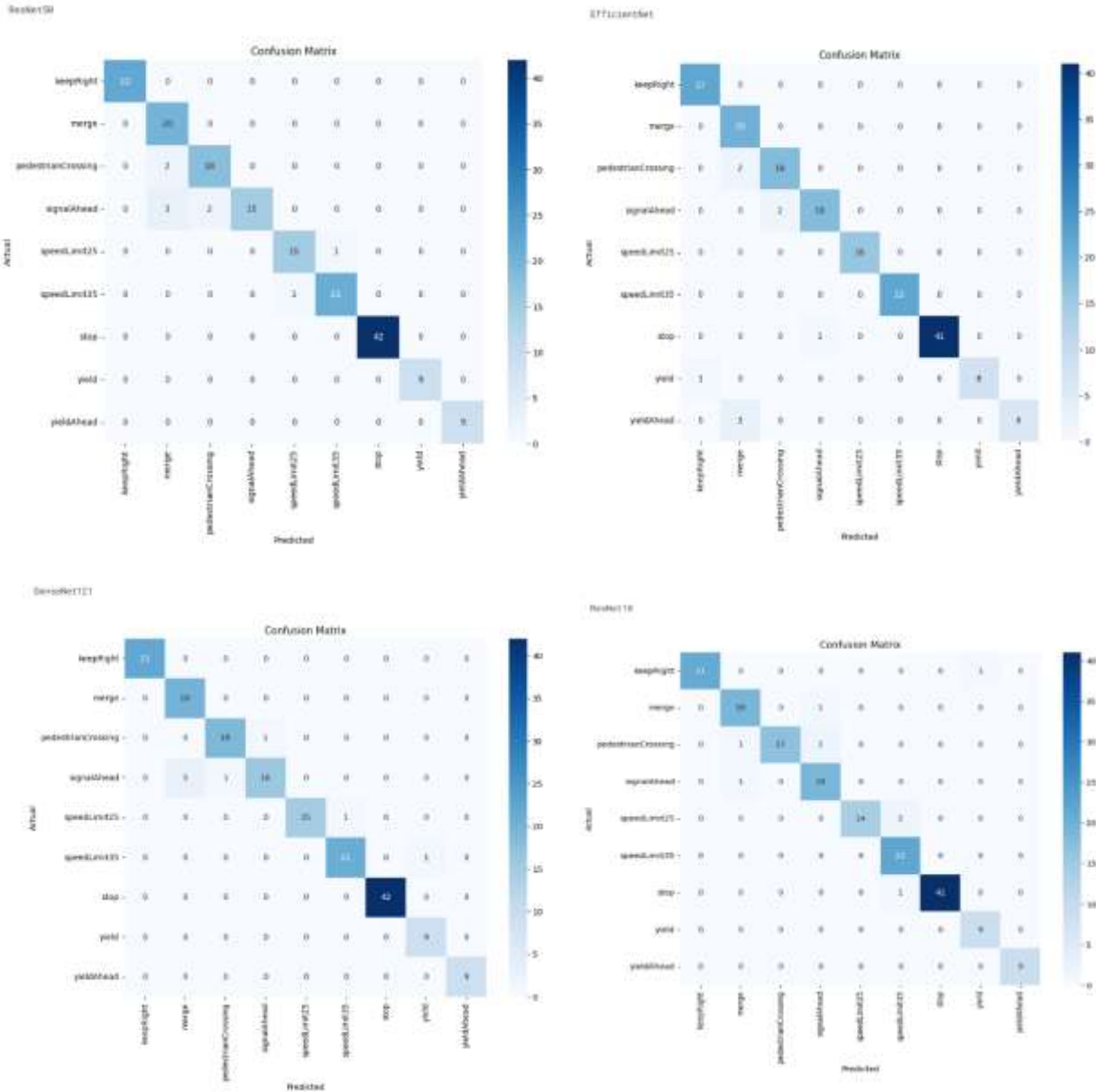


Figure 4.2: Confusion Matrices of Individual Deep Learning Models

Figure 4.2 shows the confusion matrices of the individual deep learning models demonstrate strong classification performance for traffic sign recognition, with most predictions concentrated along the diagonal, indicating accurate classification of the majority of classes. ResNet50, EfficientNet, and DenseNet121 show high recognition capability with very few misclassifications. Classes such as Stop, Keep Right, Merge, Yield, and Yield Ahead are classified almost perfectly across all models. Minor confusion is observed mainly between Signal Ahead and Pedestrian Crossing, as these signs possess similar visual patterns. Among

the individual models, DenseNet121 achieves the most consistent performance with excellent diagonal dominance, while EfficientNet also performs effectively with minimal classification errors. In contrast, ResNet18 shows comparatively higher misclassification rates, especially between Merge, Signal Ahead, and Pedestrian Crossing classes, indicating that its shallower architecture is less effective in extracting complex traffic sign features compared to deeper networks.

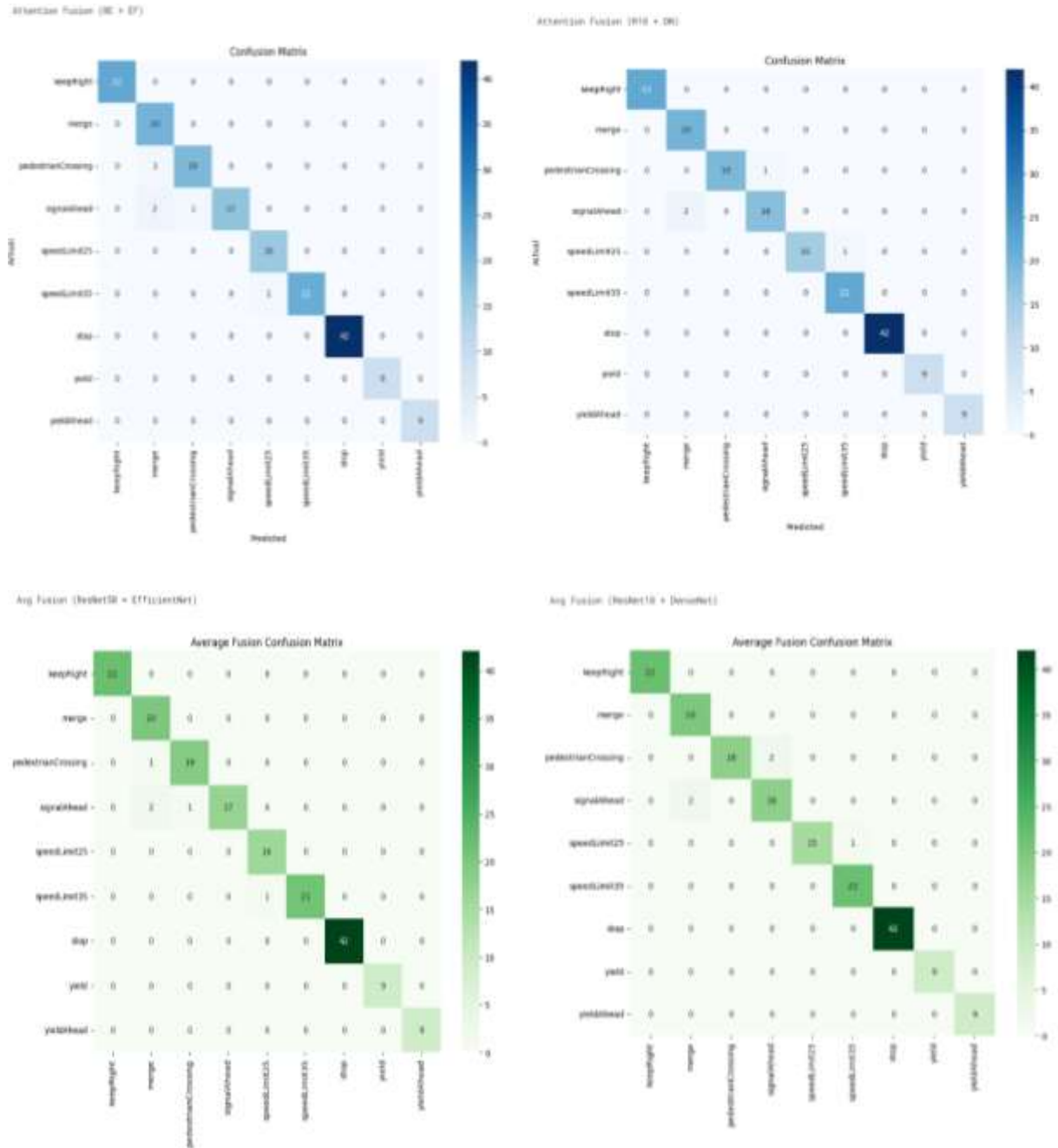


Figure 4.3: Confusion Matrices of Fusion-Based Deep Learning Models

Figure 4.3 describes the fusion-based confusion matrices further improve classification accuracy by combining complementary features from multiple models. Attention Fusion models, particularly the combinations of ResNet50 + EfficientNet and ResNet18 + DenseNet121, significantly reduce classification errors and improve class separability. These models demonstrate stronger robustness and better generalization compared to individual architectures. Most classes are predicted correctly with nearly perfect diagonal values, showing the effectiveness of attention-based feature fusion in enhancing recognition performance. Similarly, the Average Fusion models also improve overall classification accuracy when compared with standalone networks by reducing misclassifications and increasing prediction consistency. However, slight confusion still remains for visually similar classes such as Signal Ahead and Pedestrian Crossing. Overall, the confusion matrix analysis confirms that fusion strategies, especially attention-based fusion approaches, provide superior traffic sign recognition performance and enhance the reliability of the proposed system.

Table 4.3: Performance Comparison of Models

Model	Accuracy	Precision	Recall	F1 Score
ResNet18	0.9778	0.9795	0.9778	0.9777
ResNet50	0.9500	0.9520	0.9500	0.9503
EfficientNet-B0	0.9722	0.9744	0.9722	0.9726
DenseNet121	0.9556	0.9568	0.9556	0.9556
MobileNet-V2	0.9667	0.9698	0.9667	0.9671
Avg Fusion (ResNet50 + EfficientNet-B0)	0.9722	0.9735	0.9722	0.9723
Avg Fusion (ResNet18 + DenseNet121)	0.9778	0.9782	0.9778	0.9779
Attention Fusion (ResNet50 + EfficientNet-B0)	0.9833	0.9848	0.9833	0.9834
Attention Fusion (ResNet18 + DenseNet121)	0.9833	0.9836	0.9833	0.9833

Table 4.3 presents a comparison of different deep learning models and fusion techniques based on evaluation metrics such as accuracy, precision, recall, and F1-score. Among the individual models, ResNet18 achieves the highest accuracy, followed closely by EfficientNet-B0 and MobileNet-V2, indicating strong classification performance. DenseNet121 and ResNet50 show comparatively lower performance. The average fusion models improve results by

combining features from two networks, but their performance is still limited due to equal weighting of features. In contrast, the AFRNet achieve the best performance across all metrics, with the highest accuracy of 98.33% and superior precision, recall, and F1-score. This improvement demonstrates that the attention mechanism effectively selects the most relevant features from each model, leading to better classification accuracy. Overall, the results confirm that the proposed AFRNet model outperforms both individual models and traditional fusion methods, making it highly suitable for traffic sign recognition tasks.

4.4 ROC and Radar Chart Analysis

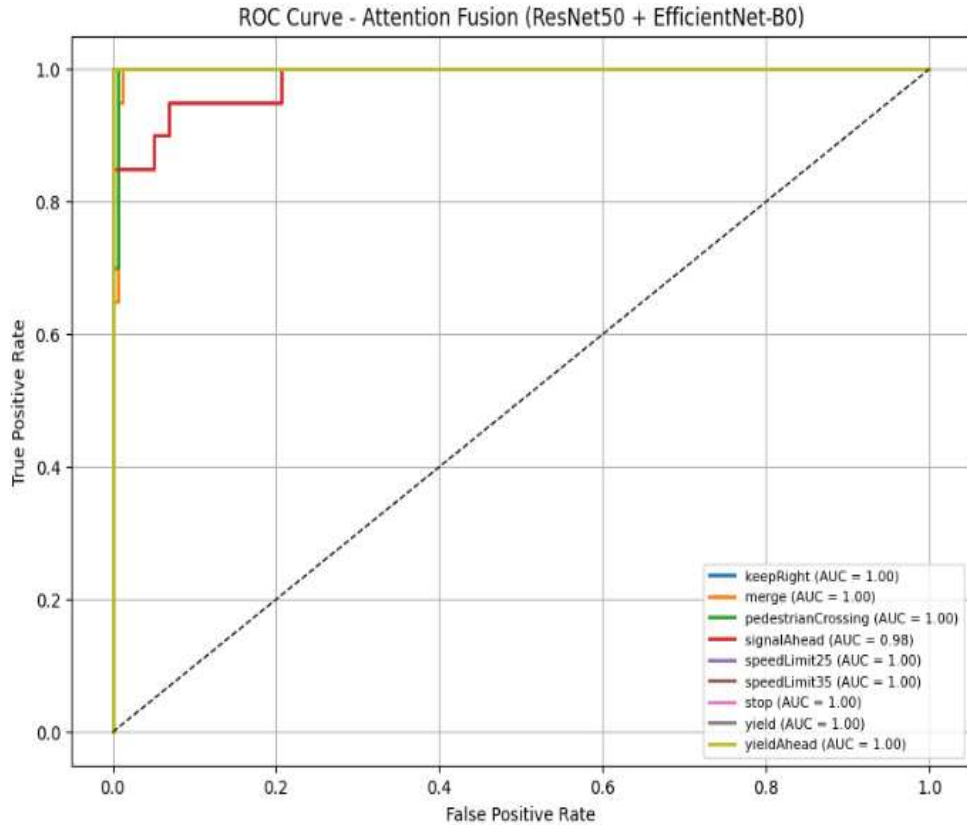


Figure 4.4: ROC curve

Figure 4.4 shows the ROC (Receiver Operating Characteristic) curve illustrates the performance of the model by plotting the True Positive Rate (TPR) against the False Positive Rate (FPR) for each traffic sign class. The curves for most classes are positioned very close to the top-left corner of the graph, indicating excellent classification performance. The Area Under the Curve (AUC) values for nearly all classes are approximately 1.0, which signifies near-perfect separability between classes. This means the model can effectively distinguish between different traffic sign categories with minimal overlap. Even for slightly challenging

classes like Signal Ahead, the AUC remains very high around 0.98, indicating strong performance despite minor misclassifications. The ROC analysis confirms that the model achieves high sensitivity while maintaining a low false positive rate, which is crucial for real-world applications such as autonomous driving systems. The consistent performance across all classes further demonstrates the robustness and reliability of the proposed model.

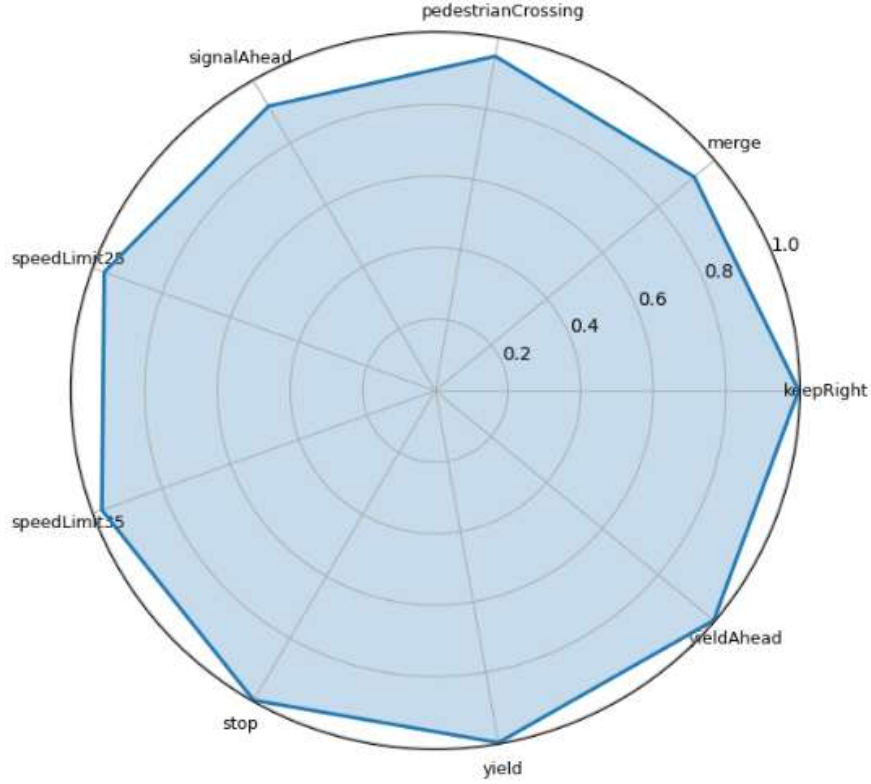


Figure 4.5: Radar Chart

Figure 4.5 shows the radar chart presents the class-wise F1-score performance of the proposed attention-based fusion model across all traffic sign categories. The F1-score, which balances precision and recall, is consistently high for all classes, indicating that the model effectively minimizes both false positives and false negatives. Classes such as Stop, Yield, Yield Ahead, and Keep Right achieve F1-scores very close to 1.0, demonstrating perfect or near-perfect classification performance. Slightly lower F1-scores are observed for classes like Merge, Pedestrian Crossing, and Signal Ahead, which can be attributed to their visual similarity and more complex patterns in real-world scenarios. Despite these minor variations, the differences are minimal, showing that the model maintains strong consistency across all classes. The uniform shape of the radar chart indicates that the model does not favor any particular class and performs equally well across all categories. This balanced performance highlights the

effectiveness of the AFRNet in capturing diverse and discriminative features from multiple CNN models.

CHAPTER-5

CONCLUSION AND FUTURE SCOPE

5. CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

The proposed traffic sign recognition system based on the attention-based fusion framework (AFRNet) effectively combines multiple CNN models such as ResNet, EfficientNet, and DenseNet to enhance feature representation and classification accuracy. Evaluated on the LISA dataset, the model demonstrates high performance across metrics like accuracy, precision, recall, and F1-score, outperforming individual and average fusion models. The confusion matrix, ROC curves, and class-wise analysis confirm consistent and reliable performance with minimal misclassification, even for visually similar signs. Overall, the AFRNet model provides a robust, scalable, and efficient solution suitable for real-world applications such as intelligent transportation systems and advanced driver assistance systems (ADAS).

5.2 Future Scope

The proposed traffic sign recognition system can be further enhanced to improve its performance and real-world applicability. One key extension is the integration of a real-time detection module to process live video streams from vehicle-mounted cameras, enabling direct use in autonomous driving and driver assistance systems. Additionally, using larger and more diverse datasets from different regions can improve the model's generalization across varying road conditions and traffic sign types. Incorporating lightweight models can reduce computational complexity, making the system suitable for deployment on edge devices. Future work may also include the use of explainable AI techniques to improve model interpretability and the integration of temporal information from video sequences to enhance recognition accuracy. Overall, these improvements can make the system more robust, efficient, and adaptable for intelligent transportation applications.

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